

GATE 2027 Electrical Engineering

Independent Practice Mock Set 01 — Answer Key and Solutions

INDEPENDENT PRACTICE MATERIAL — NOT AN OFFICIAL GATE PAPER

Compact answer key

1. B · **2.** B · **3.** A, C, D · **4.** A · **5.** 88.33 · **6.** C · **7.** A, B · **8.** B · **9.** A, D · **10.** A, D
11. C · **12.** A, B, C · **13.** B · **14.** A, B, D · **15.** B · **16.** A, B, C, D · **17.** B · **18.** A, B, D · **19.** 0.324 · **20.** C
21. B · **22.** A, C, D · **23.** 4 · **24.** 6.8 · **25.** A · **26.** A, B, C · **27.** B · **28.** C · **29.** 8.333 · **30.** A, B, C, D
31. 7.057 · **32.** 1.5 · **33.** C · **34.** B · **35.** A, B, D · **36.** 12.8 · **37.** 3.5 · **38.** B · **39.** C · **40.** 1.333
41. A, C, D · **42.** C · **43.** 0.25 · **44.** 3 · **45.** C · **46.** A, B, C · **47.** 27 · **48.** C · **49.** 0.667 · **50.** A
51. A, B, C · **52.** 8 · **53.** B · **54.** C · **55.** 9.994 · **56.** A, B, C · **57.** B · **58.** 359.5 · **59.** 2 · **60.** C
61. A, B, C · **62.** 0.0838 · **63.** C · **64.** 320 · **65.** C

Detailed solutions

Section A — General Aptitude (10 questions, 15 marks)

Q1. [1 mark; MCQ]

In a vocabulary-in-context review, the committee described its approval as “qualified” because two safety tests were still unresolved. Here, the word “qualified” most nearly means

- **A.** unconditional
- **B.** limited by conditions
- **C.** professionally trained
- **D.** officially certified

Answer: B

Solution:

The unresolved tests restrict the approval. In this context, “qualified” means conditional or limited rather than unconditional, trained, or formally certified. Final answer: B

Q2. [2 marks; MCQ]

Read the following passage and answer the reading-comprehension question. An office tested a flexible starting-time policy for four weeks. Among the 120 employees who voluntarily submitted travel data, the recorded mean commute duration fell from 52 minutes before the pilot to 46 minutes during it. No travel data were collected from nonparticipants. Which conclusion is best supported?

- **A.** The policy reduced the commute duration of every employee.
- **B.** Among the reporting employees, the recorded mean commute duration was 6 minutes lower during the pilot.
- **C.** The nonparticipants experienced the same mean reduction as the participants.
- **D.** The pilot proves that traffic volume across the entire city decreased.

Answer: B

Solution:

The stated measurements establish only that the mean for the reporting group changed from 52 to 46 minutes, a reduction of 6 minutes. Voluntary participation prevents conclusions about every employee, nonparticipants, or citywide traffic. Final answer: B

Q3. [2 marks; MSQ]

Read the following passage and select all supported statements. A fault-prediction model scored 68% on a fixed held-out sample from Plant B. After recalibration using a separate training sample from Plant B, it scored 81% on that same held-out sample. No post-recalibration test was performed at Plant A.

- **A.** The measured accuracy on the held-out Plant B sample increased by 13 percentage points.
- **B.** Recalibration guarantees improved accuracy on every future case from Plant B.
- **C.** The post-recalibration accuracy at Plant A is unknown from the given information.
- **D.** Relative to the original 68% score, the measured increase is approximately 19.1%.

Answer: A, C, D

Solution:

The percentage-point increase is $81 - 68 = 13$. The relative increase is $\frac{13}{68} \times 100\% = 19.1176\%$, approximately 19.1%. No new Plant A result is supplied, and one held-out result cannot guarantee every future outcome. Final answer: A, C, D

Q4. [1 mark; MCQ]

For a narrative-sequencing task, arrange the following sentences into the most coherent order. P: Only then could results from repeated tests be compared. Q: The initial instruction merely asked the team to improve reliability. R: The team therefore defined failure as more than two interruptions per 100 operating hours. S: Without a numerical criterion, however, success could not be judged.

- **A.** Q-S-R-P
- **B.** Q-R-S-P
- **C.** S-Q-P-R
- **D.** R-Q-S-P

Answer: A

Solution:

Q introduces the vague instruction. S explains its defect, R supplies the resulting measurable definition, and P states what that definition then enables. The coherent order is Q-S-R-P. Final answer: A

Q5. [1 mark; NAT]

A data-interpretation table gives inspection results for three quarters.

Quarter	Accepted	Rejected
Q1	84	16
Q2	108	12
Q3	126	14

Enter the overall acceptance percentage, rounded to two decimal places.

Answer: 88.33

Solution:

The total accepted count is $84 + 108 + 126 = 318$, while the total inspected count is $318 + (16 + 12 + 14) = 360$. Thus the overall acceptance percentage is $\frac{318}{360} \times 100\% = 88.333\ldots\%$, which rounds to 88.33%. Final answer: 88.33

Q6. [2 marks; MCQ]

In a permutations-and-combinations problem, six distinct jobs A, B, C, D, E and F are arranged in a sequence. Job A must occur before job B, and jobs C and D must not be adjacent. The number of valid sequences is

- **A.** 120
- **B.** 180
- **C.** 240
- **D.** 360

Answer: C

Solution:

Exactly half of the $6! = 720$ sequences have A before B, giving 360. If C and D are adjacent, treating them as one block gives $2(5!) = 240$ sequences; by symmetry, half of these, namely 120, have A before B. Therefore the valid count is $360 - 120 = 240$. Final answer: C

Q7. [1 mark; MSQ]

For a logic-deduction task, consider the premises: (i) every relay trip is logged; (ii) some logged events are manually generated; and (iii) no manually generated event is a fault. Select all conclusions that must follow.

- **A.** No manually generated event is a fault.
- **B.** Some logged events are not faults.
- **C.** Some relay trips are not faults.
- **D.** Every logged event is a relay trip.

Answer: A, B

Solution:

Statement A repeats premise (iii). By premise (ii), at least one logged event is manually generated; premise (iii) makes that event a non-fault, so B follows. Nothing states that any relay trip is a non-fault, and premise (i) does not justify its converse. Final answer: A, B

Q8. [2 marks; MCQ]

In an analytical-aptitude scheduling problem, four inspections P, Q, R and S occupy positions 1 through 4, one per position. P is before Q, R is immediately after P, and S is not fourth. Which statement must be true?

- **A.** P is first.
- **B.** Q is fourth.
- **C.** R is second.
- **D.** S is first.

Answer: B

Solution:

If P is first, R is second, and the condition on S forces the order P-R-S-Q. If P is second, R is third, forcing S-P-R-Q. P cannot be third because Q would then have no later position. Both feasible orders place Q fourth, while none of the other statements holds in both. Final answer: B

Q9. [2 marks; MSQ]

For a logic-deduction task, use these premises: (i) every device that passed test T was calibrated; (ii) some calibrated devices failed test S; and (iii) no device that failed test S was shipped. Select all conclusions that necessarily follow.

- **A.** Some calibrated devices were not shipped.
- **B.** Every shipped device passed test T.
- **C.** Some devices that passed test T were shipped.
- **D.** No uncalibrated device passed test T.

Answer: A, D

Solution:

The devices asserted in premise (ii) are calibrated and failed S; premise (iii) makes them not shipped, so A follows. Premise (i) is equivalent to saying that no uncalibrated device passed T, so D follows. The premises do not establish B or the existence claim in C. Final answer: A, D

Q10. [1 mark; MSQ]

In a three-dimensional spatial-apptitude problem, a cube has distinct face labels P, Q, R, S, T and U. The opposite face pairs are P-U, Q-T and R-S. Select all triples that can meet at a single vertex.

- **A.** P, Q, R
- **B.** P, U, T
- **C.** Q, R, S
- **D.** P, T, S

Answer: A, D

Solution:

A vertex contains exactly one face from each opposite pair. P-Q-R and P-T-S each select one label from P-U, one from Q-T, and one from R-S, so they are possible. P-U-T contains the opposite pair P-U, and Q-R-S contains the opposite pair R-S. Final answer: A, D

Section B — Engineering Mathematics (8 questions, 13 marks)**Q11. [1 mark; MCQ]**

Determine the largest eigenvalue of the matrix $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$.

- **A.** 1
- **B.** 2
- **C.** 3
- **D.** 4

Answer: C

Solution:

The characteristic equation is $\det(A - \lambda I) = 0$, so $(2 - \lambda)^2 - 1 = 0$, or $\lambda^2 - 4\lambda + 3 = 0$. Thus the eigenvalues are 1 and 3, and the largest is 3. Final answer: C

Q12. [2 marks; MSQ]

Let P be a real symmetric idempotent matrix, i.e. $P^T = P$ and $P^2 = P$. Which of the following statements are necessarily true?

- **A.** Every eigenvalue of P is either 0 or 1.
- **B.** P is positive semidefinite.
- **C.** $\text{tr}(P) = \text{rank}(P)$.
- **D.** $\det(P) = 1$.

Answer: A, B, C

Solution:

If $Pv = \lambda v$, then $P^2v = \lambda^2v$, but $P^2 = P$, so $\lambda^2 = \lambda$ and $\lambda \in \{0, 1\}$. Since P is symmetric, it is orthogonally diagonalizable with nonnegative eigenvalues, hence positive semidefinite. The trace is the sum of eigenvalues and therefore equals the number of unit eigenvalues, which is the rank. The determinant need not be 1; it is 0 whenever P has a zero eigenvalue. Final answer: A, B, C

Q13. [1 mark; MCQ]

Evaluate the exponential limit $\lim_{x \rightarrow 0} \frac{e^{2x} - 1 - 2x}{x^2}$.

- A. 1
- B. 2
- C. 4
- D. Does not exist

Answer: B

Solution:

Using $e^{2x} = 1 + 2x + \frac{(2x)^2}{2} + O(x^3) = 1 + 2x + 2x^2 + O(x^3)$, the numerator is $2x^2 + O(x^3)$. Dividing by x^2 and taking the limit gives 2. Final answer: B

Q14. [2 marks; MSQ]

Consider the second-order linear ordinary differential equation $y'' + 4y' + 3y = 0$. Which of the following functions are solutions?

- A. e^{-x}
- B. e^{-3x}
- C. e^{-2x}
- D. $2e^{-x} - 5e^{-3x}$

Answer: A, B, D

Solution:

The characteristic equation is $r^2 + 4r + 3 = (r + 1)(r + 3) = 0$, giving roots -1 and -3 . Hence every solution has the form $C_1 e^{-x} + C_2 e^{-3x}$. Therefore A, B and D are solutions, while e^{-2x} is not. Final answer: A, B, D

Q15. [1 mark; MCQ]

Evaluate $(1 + i)^4$.

- A. 4
- B. -4
- C. $4i$
- D. $-4i$

Answer: B

Solution:

$(1 + i)^2 = 1 + 2i + i^2 = 2i$. Therefore $(1 + i)^4 = (2i)^2 = -4$. Final answer: B

Q16. [2 marks; MSQ]

Let a random variable X have finite mean μ and variance σ^2 . Which statements are necessarily true?

- **A.** $E[X - \mu] = 0$
- **B.** $E[X^2] = \sigma^2 + \mu^2$
- **C.** $\text{Var}(aX + b) = a^2\sigma^2$ for constants a, b
- **D.** $E[(X - \mu)^2] = \sigma^2$

Answer: A, B, C, D

Solution:

By definition $\mu = E[X]$, so $E[X - \mu] = 0$. Also $\sigma^2 = E[X^2] - \mu^2$, giving $E[X^2] = \sigma^2 + \mu^2$. Adding a constant does not change variance and scaling by a scales variance by a^2 . Finally, variance is defined as $E[(X - \mu)^2]$. Hence all four statements are true. Final answer: A, B, C, D

Q17. [2 marks; MCQ]

Let C be the positively oriented contour $|z| = 2$. Evaluate $\oint_C \frac{z^2 + 1}{z(z - 1)} dz$.

- **A.** 0
- **B.** $2\pi i$
- **C.** $4\pi i$
- **D.** $-2\pi i$

Answer: B

Solution:

The poles $z = 0$ and $z = 1$ both lie inside C . Their residues are $\text{Res}(f, 0) = -1$ and $\text{Res}(f, 1) = 2$, so their sum is 1. By the residue theorem, the contour integral is $2\pi i$. Final answer: B

Q18. [2 marks; MSQ]

For the planar vector field $\mathbf{F} = y\mathbf{i} - x\mathbf{j}$, which statements are true?

- **A.** $\nabla \cdot \mathbf{F} = 0$
- **B.** $\nabla \times \mathbf{F} = -2\mathbf{k}$
- **C.** $\mathbf{F}$ is conservative on $\mathbb{R}^2$
- **D.** The counter-clockwise circulation along $x^2 + y^2 = 1$ is -2π .

Answer: A, B, D

Solution:

With $P = y$ and $Q = -x$, divergence is $\partial P/\partial x + \partial Q/\partial y = 0 + 0 = 0$. The scalar curl is $\partial Q/\partial x - \partial P/\partial y = -1 - 1 = -2$, i.e. $-2\mathbf{k}$. Since the curl is nonzero, the field is not conservative. Green's theorem gives the counter-clockwise circulation as $\iint_D (-2) dA = -2\pi$ for the unit disk. Final answer: A, B, D

Section C — Electrical Engineering (47 questions, 72 marks)

Q19. [1 mark; NAT]

A $6\ \mu\text{F}$ capacitor and a $3\ \mu\text{F}$ capacitor are connected in series across an 18 V DC source. Determine the total energy stored by this capacitor network; enter the numerical value in mJ.

Answer: 0.324

Solution:

The series equivalent capacitance is $C_{\text{eq}} = \frac{(6)(3)}{6+3}\ \mu\text{F} = 2\ \mu\text{F}$. The stored energy is $W = \frac{1}{2}C_{\text{eq}}V^2 = \frac{1}{2}(2 \times 10^{-6})(18)^2 = 324 \times 10^{-6}\text{ J} = 0.324\text{ mJ}$. Final answer: 0.324

Q20. [1 mark; MCQ]

Two coupled inductors in a series-aiding network have $L_1 = 4\text{ H}$, $L_2 = 9\text{ H}$ and coefficient of coupling $k = 0.5$. Determine the equivalent inductance.

- **A.** 7 H
- **B.** 13 H
- **C.** 19 H
- **D.** 25 H

Answer: C

Solution:

The mutual inductance is $M = k\sqrt{L_1L_2} = 0.5\sqrt{(4)(9)} = 3\text{ H}$. For a series-aiding connection, $L_{\text{eq}} = L_1 + L_2 + 2M = 4 + 9 + 6 = 19\text{ H}$. Final answer: C

Q21. [2 marks; MCQ]

At the input port of a linear network, a $4\ \Omega$ resistor is connected from the port to ground. A dependent current source of value $0.5i_R$, where i_R is the resistor current, is also directed from the port to ground. Determine the resistance seen at the port.

- **A.** $2\ \Omega$
- **B.** $\frac{8}{3}\ \Omega$
- **C.** $4\ \Omega$
- **D.** $6\ \Omega$

Answer: B

Solution:

Apply a test voltage v at the port. The resistor current is $i_R = \frac{v}{4}$, and the dependent-source current is $0.5i_R$. Hence $i_{\text{in}} = \frac{v}{4} + 0.5\left(\frac{v}{4}\right) = \frac{3v}{8}$, so $R_{\text{in}} = \frac{v}{i_{\text{in}}} = \frac{8}{3}\ \Omega$. Final answer: B

Q22. [1 mark; MSQ]

Select every correct statement about ideal passive network elements under finite, impulse-free excitation; one or more options may be correct.

- **A.** The energy stored in a capacitor is $\frac{1}{2}Cv^2$.
- **B.** The energy stored in an inductor is Li^2 .
- **C.** A capacitor voltage cannot change instantaneously.
- **D.** An inductor current cannot change instantaneously.

Answer: A, C, D

Solution:

Capacitor energy is $\frac{1}{2}Cv^2$, whereas inductor energy is $\frac{1}{2}Li^2$; therefore option B is missing the factor $\frac{1}{2}$. With finite, impulse-free excitation, capacitor voltage and inductor current cannot change instantaneously. Therefore A, C and D are correct. Final answer: A, C, D

Q23. [2 marks; NAT]

In a resistive network, node v_1 is connected to a 12 V source through $3\ \Omega$, to ground through $6\ \Omega$, and to node v_2 through $2\ \Omega$. Node v_2 is connected to ground through $4\ \Omega$. Using KCL and node-voltage analysis, determine v_2 ; enter the numerical value in V.

Answer: 4

Solution:

KCL at v_2 gives $\frac{v_2 - v_1}{2} + \frac{v_2}{4} = 0$, hence $v_1 = 1.5v_2$. KCL at v_1 gives $\frac{v_1 - 12}{3} + \frac{v_1}{6} + \frac{v_1 - v_2}{2} = 0$. Multiplying by 6 yields $6v_1 - 3v_2 = 24$. Substitution gives $6(1.5v_2) - 3v_2 = 24$, so $v_2 = 4$ V. Final answer: 4

Q24. [2 marks; NAT]

Two non-reference node voltages satisfy $v_1 - v_2 = 5$ V because of an ideal voltage source between them. A $2\ \Omega$ branch connects v_1 to ground, a $3\ \Omega$ branch connects v_2 to ground, and a 4 A current source injects current into v_2 . Using supernode KCL, determine v_1 ; enter the numerical value in V.

Answer: 6.8

Solution:

KCL for the supernode is $\frac{v_1}{2} + \frac{v_2}{3} = 4$. The voltage-source constraint is $v_1 = v_2 + 5$. Substituting and multiplying by 6 gives $3(v_2 + 5) + 2v_2 = 24$, so $v_2 = 1.8$ V and $v_1 = 6.8$ V. Final answer: 6.8

Q25. [2 marks; MCQ]

Clockwise mesh currents I_1 and I_2 in a resistor network satisfy $5I_1 - 2I_2 = 8$ and $-2I_1 + 4I_2 = 2$. Determine the current in their common branch, directed with I_1 .

- **A.** 0.625 A
- **B.** 1.625 A
- **C.** 2.250 A
- **D.** 3.875 A

Answer: A

Solution:

Solving the mesh-current equations gives $I_1 = \frac{36}{16} = 2.25$ A and $I_2 = \frac{26}{16} = 1.625$ A. The common-branch current in the I_1 direction is $I_1 - I_2 = 0.625$ A. Final answer: A

Q26. [1 mark; MSQ]

Select every correct statement about node-voltage analysis of a connected electrical network; one or more options may be correct.

- **A.** One node is selected as the reference node.
- **B.** An ideal voltage source between two non-reference nodes can be handled using a supernode.
- **C.** For n nodes, at most $n - 1$ independent KCL node equations are required.
- **D.** The method cannot be used when dependent sources are present.

Answer: A, B, C

Solution:

Node-voltage analysis selects a reference, uses supernodes for ideal voltage sources between unknown nodes, and requires at most $n - 1$ independent KCL equations. Dependent sources are permitted when their controlling relations are included. Thus A, B and C are correct. Final answer: A, B, C

Q27. [2 marks; MCQ]

An 18 V ideal source feeds a $3\ \Omega$ series resistor and an output node; a $6\ \Omega$ resistor connects that node to ground. A $4\ \Omega$ load is then connected from the output node to ground. Determine the load current using the Thevenin equivalent of the source network.

- **A.** 1 A
- **B.** 2 A
- **C.** 3 A
- **D.** 4 A

Answer: B

Solution:

The open-circuit voltage is $V_{th} = 18 \left(\frac{6}{3+6} \right) = 12$ V. With the independent source deactivated, $R_{th} = 3 \parallel 6 = 2\ \Omega$. Therefore $I_L = \frac{12}{2+4} = 2$ A. Final answer: B

Q28. [1 mark; MCQ]

A Norton network consists of a 3 A current source in parallel with $6\ \Omega$. When a $3\ \Omega$ load is connected across the network, determine the load current.

- **A.** 0.5 A
- **B.** 1 A
- **C.** 2 A
- **D.** 3 A

Answer: C

Solution:

By current division, the 3 A Norton current splits between $6\ \Omega$ and $3\ \Omega$. The load current is $I_L = 3 \left(\frac{6}{6+3} \right) = 2\text{ A}$. Final answer: C

Q29. [2 marks; NAT]

A sinusoidal network has Thevenin RMS voltage $V_{th} = 10\text{ V}$ and Thevenin impedance $Z_{th} = (3 + j4)\ \Omega$. A conjugately matched load is connected. Determine the maximum average load power; enter the numerical value in W.

Answer: 8.333

Solution:

For conjugate matching, $Z_L = Z_{th}^* = (3 - j4)\ \Omega$. The maximum average load power is $P_{max} = \frac{|V_{th}|^2}{4R_{th}} = \frac{10^2}{4(3)} = 8.333\text{ W}$. Final answer: 8.333

Q30. [1 mark; MSQ]

Select every correct statement about superposition in a linear electrical network; one or more options may be correct.

- **A.** An independent ideal voltage source is replaced by a short circuit when deactivated.
- **B.** An independent ideal current source is replaced by an open circuit when deactivated.
- **C.** Dependent sources remain active.
- **D.** The total power cannot in general be obtained by adding the powers caused by the individual sources.

Answer: A, B, C, D

Solution:

Superposition deactivates only independent sources: ideal voltage sources become shorts and ideal current sources become opens. Dependent sources remain because they are part of the network model. Voltages and currents superpose, but power is nonlinear, so the separate-source powers do not generally add. All four statements are correct. Final answer: A, B, C, D

Q31. [2 marks; NAT]

In a first-order RC network, the capacitor voltage is initially 2 V and its steady-state value after switching is 10 V. The resistance is $2\text{ k}\Omega$ and the capacitance is $100\text{ }\mu\text{F}$. Determine the capacitor voltage one time constant after switching; enter the numerical value in V.

Answer: 7.057

Solution:

The time constant is $\tau = RC = (2000)(100 \times 10^{-6}) = 0.2\text{ s}$. At $t = \tau$, $v_C(\tau) = 10 + (2 - 10)e^{-1} = 10 - \frac{8}{e} = 7.057\text{ V}$. Final answer: 7.057

Q32. [2 marks; NAT]

A 12 V DC source is applied at $t = 0$ to an initially unenergized series RL network with resistance $R = 4\text{ }\Omega$ and inductance $L = 2\text{ H}$. Determine the inductor current at $t = (0.5 \ln 2)\text{ s}$; enter the numerical value in A.

Answer: 1.5

Solution:

The final current is $I_\infty = \frac{12}{4} = 3\text{ A}$, and the time constant is $\tau = \frac{L}{R} = \frac{2}{4} = 0.5\text{ s}$. Here $\frac{t}{\tau} = \ln 2$, so $i_L(t) = 3(1 - e^{-\ln 2}) = 3(1 - \frac{1}{2}) = 1.5\text{ A}$. Final answer: 1.5

Q33. [1 mark; MCQ]

A series RLC network has inductance $L = 50\text{ mH}$ and capacitance $C = 20\text{ }\mu\text{F}$. Determine its undamped resonant angular frequency.

- **A.** 100 rad s^{-1}
- **B.** 500 rad s^{-1}
- **C.** 1000 rad s^{-1}
- **D.** 5000 rad s^{-1}

Answer: C

Solution:

The resonant angular frequency is $\omega_0 = \frac{1}{\sqrt{LC}}$. Since $LC = (50 \times 10^{-3})(20 \times 10^{-6}) = 10^{-6}$, $\omega_0 = \frac{1}{10^{-3}} = 1000\text{ rad s}^{-1}$. Final answer: C

Q34. [1 mark; MCQ]

A series RLC network has resistance $R = 20\text{ }\Omega$ and inductance $L = 0.1\text{ H}$. Determine its half-power bandwidth in angular frequency.

- **A.** 100 rad s^{-1}
- **B.** 200 rad s^{-1}
- **C.** 500 rad s^{-1}
- **D.** 2000 rad s^{-1}

Answer: B

Solution:

For series RLC resonance, the angular-frequency bandwidth is $\Delta\omega = \frac{R}{L} = \frac{20}{0.1} = 200\text{ rad s}^{-1}$. Final answer: B

Q35. [2 marks; MSQ]

A single-phase impedance load absorbs $P = 8 \text{ kW}$ at a lagging power factor of 0.8. Select every correct statement about this sinusoidal steady-state network; one or more options may be correct.

- **A.** The load reactive power is 6 kVAR.
- **B.** The apparent-power magnitude is 10 kVA.
- **C.** The load current leads the load voltage.
- **D.** A 6 kVAR shunt capacitor would correct the source power factor to unity.

Answer: A, B, D

Solution:

With $\cos \phi = 0.8$, $\tan \phi = 0.75$. Hence $Q = P \tan \phi = (8)(0.75) = 6 \text{ kVAR}$ and $|S| = \frac{P}{\cos \phi} = \frac{8}{0.8} = 10 \text{ kVA}$. A lagging load current does not lead its voltage. A 6 kVAR capacitive shunt cancels the inductive reactive power, so A, B and D are correct. Final answer: A, B, D

Q36. [2 marks; NAT]

A balanced three-phase, star-connected load has per-phase impedance $Z_{\text{ph}} = (8 + j6) \Omega$ and is supplied at a line-to-line RMS voltage $V_L = 400 \text{ V}$. Determine the total real power in kilowatts.

Answer: 12.8

Solution:

The phase-voltage magnitude is $V_{\text{ph}} = \frac{400}{\sqrt{3}} \text{ V}$, the phase-current magnitude is $I_{\text{ph}} = \frac{400/\sqrt{3}}{10} \text{ A}$, and the load power factor is $\cos \phi = \frac{8}{10} = 0.8$. Thus $P = \sqrt{3} V_L I_L \cos \phi = \sqrt{3}(400) \left(\frac{400}{10\sqrt{3}} \right) (0.8) = 12800 \text{ W} = 12.8 \text{ kW}$. Final answer: 12.8

Q37. [2 marks; NAT]

A two-port network has $z_{11} = 4 \Omega$, $z_{12} = z_{21} = 2 \Omega$, and $z_{22} = 5 \Omega$. Port 2 is terminated by a load $Z_L = 3 \Omega$. Determine the input impedance; enter the numerical value in Ω .

Answer: 3.5

Solution:

For a z -parameter two-port terminated by Z_L , the input impedance is $Z_{\text{in}} = z_{11} - \frac{z_{12}z_{21}}{z_{22} + Z_L}$. Therefore $Z_{\text{in}} = 4 - \frac{(2)(2)}{5+3} = 4 - 0.5 = 3.5 \Omega$. Final answer: 3.5

Q38. [2 marks; MCQ]

A balanced three-phase, delta-connected load has branch-impedance magnitude $|Z_{\Delta}| = 12\ \Omega$ and lagging phase angle 30° . It is supplied at a line-to-line RMS voltage $V_L = 240\text{ V}$. Determine the approximate total real power.

- A. 7.20 kW
- B. 12.47 kW
- C. 14.40 kW
- D. 24.94 kW

Answer: B

Solution:

For a delta load, $V_{\text{ph}} = V_L = 240\text{ V}$, so $I_{\text{ph}} = \frac{240}{12} = 20\text{ A}$. The total real power is $P = 3V_{\text{ph}}I_{\text{ph}}\cos 30^\circ = 3(240)(20)\cos 30^\circ = 12470.8\text{ W} \approx 12.47\text{ kW}$. Final answer: B

Q39. [1 mark; MCQ]

The continuous-time signal $x(t) = \cos(4t) + \sin(6t)$ is periodic. Determine its fundamental period.

- A. $\frac{\pi}{3}\text{ s}$
- B. $\frac{\pi}{2}\text{ s}$
- C. $\pi\text{ s}$
- D. $2\pi\text{ s}$

Answer: C

Solution:

The angular frequencies are 4 and 6 rad/s. Their greatest common divisor is $\omega_0 = 2\text{ rad/s}$, so the fundamental period is $T_0 = \frac{2\pi}{\omega_0} = \pi\text{ s}$. Final answer: C

Q40. [1 mark; NAT]

For the discrete-time signal $x[n] = (\frac{1}{2})^n u[n]$, determine the signal energy $\sum_{n=-\infty}^{\infty} |x[n]|^2$.

Answer: 1.333

Solution:

Because $u[n]$ restricts the sequence to $n \geq 0$, the energy is $E = \sum_{n=0}^{\infty} (\frac{1}{4})^n = \frac{1}{1-1/4} = \frac{4}{3} = 1.333$ to three decimal places. Final answer: 1.333

Q41. [2 marks; MSQ]

A continuous-time system is defined by $y(t) = tx(t)$. Select all properties that this system satisfies.

- **A.** Linear
- **B.** Time invariant
- **C.** Memoryless
- **D.** Causal

Answer: A, C, D

Solution:

Superposition is preserved because multiplication by the fixed function t is linear. The output at t depends only on $x(t)$, so the system is memoryless and causal. For a shifted input, the response is $tx(t - t_0)$, whereas a shifted version of the original output is $(t - t_0)x(t - t_0)$; hence the system is time varying. Final answer: A, C, D

Q42. [1 mark; MCQ]

The triangular signal shown in the figure is $x(t) = 2 - |t|$ for $|t| \leq 2$ and zero otherwise. Determine its energy.

- **A.** $\frac{8}{3}$
- **B.** $\frac{4}{3}$
- **C.** $\frac{16}{3}$
- **D.** 8

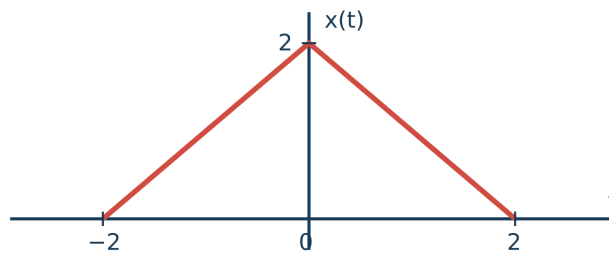


Figure: Triangular signal $x(t)$

Accessible description: Triangular signal with vertices at (-2,0), (0,2), and (2,0).

Answer: C

Solution:

Using even symmetry, $E = 2 \int_0^2 (2 - t)^2 dt = 2[-(2 - t)^3/3]_0^2 = \frac{16}{3}$. Final answer: C

Q43. [2 marks; NAT]

Let $x(t) = e^{-t}u(t)$ and $h(t) = e^{-2t}u(t)$. If $y(t) = x(t) * h(t)$, determine $y(\ln 2)$.

Answer: 0.25

Solution:

For $t \geq 0$, $y(t) = \int_0^t e^{-\tau} e^{-2(t-\tau)} d\tau = e^{-2t}(e^t - 1) = e^{-t} - e^{-2t}$. At $t = \ln 2$, this is $\frac{1}{2} - \frac{1}{4} = 0.25$.
Final answer: 0.25

Q44. [1 mark; NAT]

A real LTI system has frequency response $H(j2) = 3e^{j\pi/6}$. For input $x(t) = \cos(2t + \pi/18)$, enter the amplitude of the steady-state output sinusoid.

Answer: 3

Solution:

A sinusoid is an eigenfunction of an LTI system. Its amplitude is multiplied by $|H(j2)| = 3$, while its phase is shifted by $\pi/6$. Therefore the output amplitude is 3. Final answer: 3

Q45. [2 marks; MCQ]

For a discrete-time LTI system, $x[n] = h[n] = u[n] - u[n - 3]$. Determine the output sample $y[2]$.

- A. 1
- B. 2
- C. 3
- D. 4

Answer: C

Solution:

Both $x[n]$ and $h[n]$ equal one at $n = 0, 1, 2$. Hence $y[2] = \sum_k x[k]h[2 - k]$ has contributions for $k = 0, 1, 2$, each equal to one. Thus $y[2] = 3$. Final answer: C

Q46. [1 mark; MSQ]

The periodic waveform shown in the figure equals $+2$ for $0 \leq t < T/4$ and -2 for $T/4 \leq t < T$. Select all correct statements.

- A. Its average value is -1 .
- B. Its RMS value is 2.
- C. Its average power is 4.
- D. Its DC component is zero.

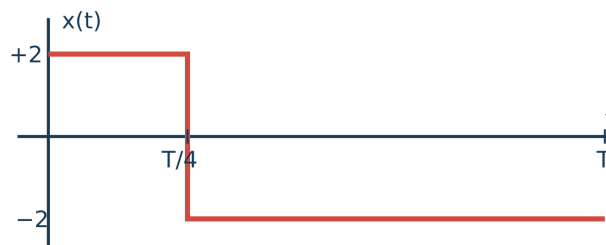


Figure: Two-level periodic waveform

Accessible description: One period of a waveform at $+2$ for the first quarter-period and -2 for the remaining three quarters.

Answer: A, B, C

Solution:

The average is $2(1/4) - 2(3/4) = -1$. Since $x^2(t) = 4$ throughout the period, the RMS value is $\sqrt{4} = 2$ and the average power is 4. The nonzero average means the DC component is not zero. Final answer: A, B, C

Q47. [2 marks; NAT]

The periodic input to a real LTI system is $x(t) = 2 + 4 \cos(\omega_0 t) + 3 \sin(2\omega_0 t)$. The frequency-response magnitudes are $|H(0)| = \frac{1}{2}$, $|H(j\omega_0)| = 1$, and $|H(j2\omega_0)| = 2$. Determine the average output power.

Answer: 27

Solution:

The output DC value is $2(1/2) = 1$. The two sinusoidal output amplitudes are 4 and 6. Orthogonality of distinct harmonics gives $P_y = 1^2 + \frac{4^2}{2} + \frac{6^2}{2} = 1 + 8 + 18 = 27$. Final answer: 27

Q48. [1 mark; MCQ]

The signal $x(t) = e^{-2t}u(t) - e^{3t}u(-t)$ has a bilateral Laplace transform. Which region of convergence applies?

- A. $\text{Re}(s) > 3$
- B. $\text{Re}(s) < -2$
- C. $-2 < \text{Re}(s) < 3$
- D. $\text{Re}(s) > -2$

Answer: C

Solution:

The right-sided term requires $\text{Re}(s) > -2$. The left-sided term requires $\text{Re}(s) < 3$. Their intersection is $-2 < \text{Re}(s) < 3$. Final answer: C

Q49. [2 marks; NAT]

A causal signal has Laplace transform $X(s) = \frac{s+4}{s(s+2)(s+3)}$. Determine its final value $\lim_{t \rightarrow \infty} x(t)$.

Answer: 0.667

Solution:

All poles of $sX(s)$ lie in the open left half-plane, so the final-value theorem applies. Hence $x(\infty) = \lim_{s \rightarrow 0} sX(s) = \frac{4}{(2)(3)} = \frac{2}{3} = 0.667$ to three decimal places. Final answer: 0.667

Q50. [1 mark; MCQ]

For $X(z) = \frac{z}{z-0.5} - \frac{z}{z-2}$ with region of convergence $|z| > 2$, determine $x[n]$.

- A. $[(0.5)^n - 2^n]u[n]$
- B. $[(0.5)^n - 2^n]u[-n-1]$
- C. $(0.5)^n u[n] + 2^n u[-n-1]$
- D. $-(0.5)^n u[-n-1] - 2^n u[n]$

Answer: A

Solution:

The stated region lies outside both poles, so both components are right sided. Using $\frac{z}{z-a} \leftrightarrow a^n u[n]$ gives $x[n] = (0.5)^n u[n] - 2^n u[n]$. Final answer: A

Q51. [2 marks; MSQ]

Consider $H(z) = \frac{1-0.5z^{-1}}{(1-0.25z^{-1})(1-2z^{-1})}$. Select all correct statements about possible regions of convergence.

- A. For $|z| > 2$, the system is causal but unstable.
- B. For $0.25 < |z| < 2$, the system is stable and two sided.
- C. For $|z| < 0.25$, the system is anti-causal and unstable.
- D. The system is FIR for every possible region of convergence.

Answer: A, B, C

Solution:

An outer ROC gives a right-sided causal impulse response, but it excludes the unit circle and is unstable. The annular ROC produces a two-sided response and includes the unit circle, so it is stable. The inner ROC gives a left-sided anti-causal response and excludes the unit circle, so it is unstable. The two uncanceled poles make the response IIR. Final answer: A, B, C

Q52. [1 mark; NAT]

The magnitude spectrum shown in the figure is nonzero only over $2 \leq |f| \leq 4$ kHz. Using the conventional baseband sampling theorem, enter the minimum sampling frequency in kHz that guarantees no aliasing.

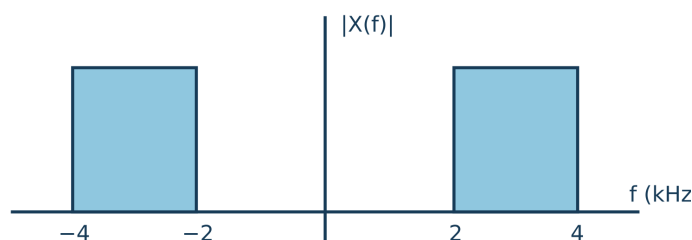


Figure: Magnitude spectrum $|X(f)|$

Accessible description: Two rectangular spectral bands from -4 to -2 kHz and from 2 to 4 kHz.

Answer: 8

Solution:

The highest frequency present is 4 kHz. The conventional Nyquist criterion requires $f_s \geq 2f_{\max} = 8$ kHz. Final answer: 8

Q53. [2 marks; MCQ]

The signal $x(t) = \cos(2\pi 1000t) + \cos(2\pi 2600t)$ is sampled at 4 kHz. The samples are passed through an ideal reconstruction low-pass filter with passband $0 \leq f \leq 2$ kHz. Which two positive-frequency components appear in the reconstructed signal?

- A. 0.6 kHz and 1.0 kHz
- B. 1.0 kHz and 1.4 kHz
- C. 1.0 kHz and 2.0 kHz
- D. 1.4 kHz and 2.6 kHz

Answer: B

Solution:

The 1 kHz component lies below Nyquist and is unchanged. The 2.6 kHz component folds about 4 kHz to $|2.6 - 4| = 1.4$ kHz. Both 1.0 and 1.4 kHz pass through the reconstruction filter. Final answer: B

Q54. [1 mark; MCQ]

A point charge Q is located at the centre of a closed spherical surface. Which value equals $\oint_S \mathbf{D} \cdot d\mathbf{S}$?

- A. 0
- B. $Q/(4\pi)$
- C. Q
- D. Q/ϵ_0

Answer: C

Solution:

Gauss law for electric flux density states $\oint_S \mathbf{D} \cdot d\mathbf{S} = Q_{\text{enclosed}}$. The sphere encloses the complete point charge, so the integral equals Q . Final answer: C

Q55. [1 mark; NAT]

An infinite line charge in free space has uniform density 55.6 nC/m. Determine the electric-field magnitude at radial distance 0.1 m; enter the value in kV/m. Use $\epsilon_0 = 8.854 \times 10^{-12}$ F/m.

Answer: 9.994

Solution:

For an infinite line charge, $E = \frac{\lambda}{2\pi\epsilon_0\rho}$. Substitution gives $E = \frac{55.6 \times 10^{-9}}{2\pi(8.854 \times 10^{-12})(0.1)} = 9.994 \times 10^3$ V/m = 9.994 kV/m. Final answer: 9.994

Q56. [2 marks; MSQ]

The figure shows a charge-free boundary between dielectric 1 with $\epsilon_{r1} = 2$ and dielectric 2 with $\epsilon_{r2} = 5$. In medium 1, the electric field has normal component 3 V/m and tangential component 4 V/m . Select all correct statements for medium 2.

- A. $D_{2n} = D_{1n}$
- B. $E_{2t} = E_{1t}$
- C. $E_{2n} = 1.2 \text{ V/m}$
- D. $|\mathbf{E}_2| = 5 \text{ V/m}$

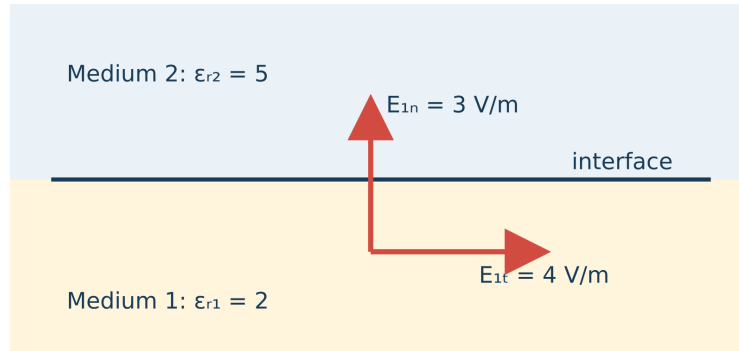


Figure: Charge-free dielectric interface

Accessible description: Horizontal dielectric interface with a normal axis, a tangential axis, and electric-field components in medium 1.

Answer: A, B, C

Solution:

With no free surface charge, the normal component of $\mathbf{D}$ is continuous, so $\epsilon_1 E_{1n} = \epsilon_2 E_{2n}$ and $E_{2n} = (2/5)(3) = 1.2 \text{ V/m}$. The tangential component of $\mathbf{E}$ is continuous, so $E_{2t} = 4 \text{ V/m}$. Therefore $|\mathbf{E}_2| = \sqrt{1.2^2 + 4^2} \neq 5 \text{ V/m}$. Final answer: A, B, C

Q57. [1 mark; MCQ]

Two concentric conducting spheres of radii a and b , with $b > a$, are separated by a dielectric of permittivity ϵ . Neglect fringing. Which expression gives the capacitance?

- A. $4\pi\epsilon(a + b)$
- B. $4\pi\epsilon \frac{ab}{b-a}$
- C. $4\pi\epsilon \frac{b-a}{ab}$
- D. $\frac{\epsilon ab}{4\pi(b-a)}$

Answer: B

Solution:

For charge Q , the potential difference is $V = \frac{Q}{4\pi\epsilon} \left(\frac{1}{a} - \frac{1}{b} \right) = \frac{Q(b-a)}{4\pi\epsilon ab}$. Thus $C = Q/V = 4\pi\epsilon \frac{ab}{b-a}$. Final answer: B

Q58. [2 marks; NAT]

A solid sphere of radius 0.1 m carries a uniform volume charge with total charge 8 nC in free space. Determine the potential at the centre relative to the potential at the surface, in volts. Use $\epsilon_0 = 8.854 \times 10^{-12} \text{ F/m}$.

Answer: 359.5

Solution:

Inside a uniformly charged sphere, the centre-to-surface potential difference is $V(0) - V(a) = \frac{Q}{8\pi\epsilon_0 a}$.

Hence the requested value is $\frac{8 \times 10^{-9}}{8\pi(8.854 \times 10^{-12})(0.1)} = 359.5 \text{ V}$. Final answer: 359.5

Q59. [1 mark; NAT]

A parallel-plate capacitor has half of its plate area filled through the full separation with a dielectric of relative permittivity 3; the other half remains air. Neglect fringing. Enter the ratio of its capacitance to the all-air capacitance.

Answer: 2

Solution:

The two half-area regions are in parallel. Thus $C = \frac{\epsilon_0(A/2)}{d} + \frac{3\epsilon_0(A/2)}{d} = 2\frac{\epsilon_0 A}{d}$. Relative to the all-air value $\epsilon_0 A/d$, the ratio is 2. Final answer: 2

Q60. [2 marks; MCQ]

The figure shows two infinitely long parallel conductors separated by 0.2 m. The left conductor carries 5 A out of the page and the right conductor carries 10 A into the page. Determine the magnetic-flux-density magnitude at the midpoint.

- A. $10 \mu\text{T}$
- B. $15 \mu\text{T}$
- C. $30 \mu\text{T}$
- D. $45 \mu\text{T}$

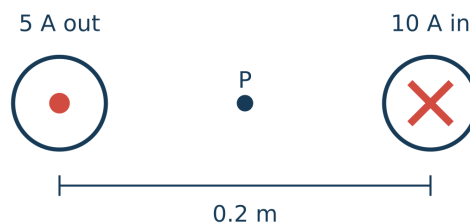


Figure: Parallel conductors and midpoint P

Accessible description: Two conductors 0.2 m apart, with 5 A out of the page on the left and 10 A into the page on the right; midpoint P is marked.

Answer: C

Solution:

The midpoint is 0.1 m from each conductor. By the right-hand rule the two fields point in the same direction there. Hence $B = \frac{\mu_0(5+10)}{2\pi(0.1)} = 30 \times 10^{-6} \text{ T} = 30 \mu\text{T}$. Final answer: C

Q61. [1 mark; MSQ]

Select all statements that are valid in magnetostatics for ordinary media without magnetic monopoles.

- **A.** $\oint_C \mathbf{H} \cdot d\mathbf{l}$ equals the free current enclosed by C .
- **B.** $\nabla \cdot \mathbf{B} = 0$.
- **C.** A free surface-current density can produce a discontinuity in tangential $\mathbf{H}$.
- **D.** The normal component of $\mathbf{B}$ may have an arbitrary discontinuity even when no magnetic monopoles exist.

Answer: A, B, C

Solution:

Ampere law gives $\oint_C \mathbf{H} \cdot d\mathbf{l} = I_{\text{free,enclosed}}$. Gauss law for magnetism gives $\nabla \cdot \mathbf{B} = 0$, which also enforces continuity of normal $\mathbf{B}$. A free surface current produces the standard tangential- $\mathbf{H}$ jump. Therefore the first three statements are correct and the fourth is false. Final answer: A, B, C

Q62. [2 marks; NAT]

A toroidal magnetic core has mean magnetic path length 0.5 m, relative permeability 1000, and a single air gap of length 1 mm. A 500-turn winding carries 0.2 A. Neglect leakage and fringing. Determine the flux density in tesla.

Answer: 0.0838

Solution:

The same flux density crosses the core and gap. The magnetic-circuit relation is $NI = \frac{B}{\mu_0\mu_r}l_c + \frac{B}{\mu_0}l_g$.

Thus $B = \frac{\mu_0 NI}{l_g + l_c/\mu_r} = \frac{(4\pi \times 10^{-7})(500)(0.2)}{0.001 + 0.5/1000} = 0.0838 \text{ T}$. Final answer: 0.0838

Q63. [1 mark; MCQ]

A single-turn loop links magnetic flux $\Phi(t) = (4t^2 + 2t) \text{ mWb}$. Determine the magnitude of the induced emf at $t = 0.5 \text{ s}$.

- **A.** 2 mV
- **B.** 4 mV
- **C.** 6 mV
- **D.** 8 mV

Answer: C

Solution:

Faraday law gives $|e| = |d\Phi/dt|$. Here $d\Phi/dt = (8t + 2) \text{ mWb/s}$, so at $t = 0.5 \text{ s}$ the magnitude is 6 mV. Final answer: C

Q64. [2 marks; NAT]

A conducting rod of length 0.5 m moves at 4 m/s perpendicular to a uniform magnetic field of 0.8 T. The rod completes a circuit of total resistance $2\ \Omega$. Neglect friction. Enter the external force required to maintain constant speed, in millinewtons.

Answer: 320

Solution:

The motional emf is $e = Blv = (0.8)(0.5)(4) = 1.6\text{ V}$, so $I = e/R = 0.8\text{ A}$. The opposing magnetic force is $F = BIl = (0.8)(0.8)(0.5) = 0.32\text{ N} = 320\text{ mN}$. Final answer: 320

Q65. [2 marks; MCQ]

The cross-sectional figure shows a long solenoid of radius 0.1 m whose uniform internal magnetic flux density changes at $dB/dt = 2\text{ T/s}$. Neglect the field outside the solenoid. Determine the induced electric-field magnitude on a circular path of radius 0.2 m concentric with the solenoid.

- **A.** 0.0125 V/m
- **B.** 0.025 V/m
- **C.** 0.05 V/m
- **D.** 0.10 V/m

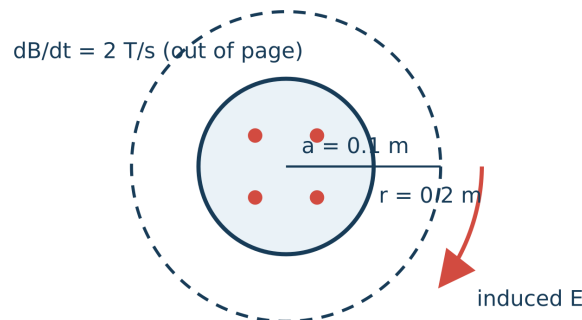


Figure: Solenoid cross-section and circular observation path

Accessible description: Concentric circles of radii 0.1 m and 0.2 m representing the solenoid cross-section and the integration path, with increasing axial magnetic field.

Answer: C

Solution:

For $r > a$, Faraday law gives $2\pi rE = \pi a^2|dB/dt|$. Therefore $E = \frac{a^2}{2r}|dB/dt| = \frac{0.1^2}{2(0.2)}(2) = 0.05\text{ V/m}$. Final answer: C